

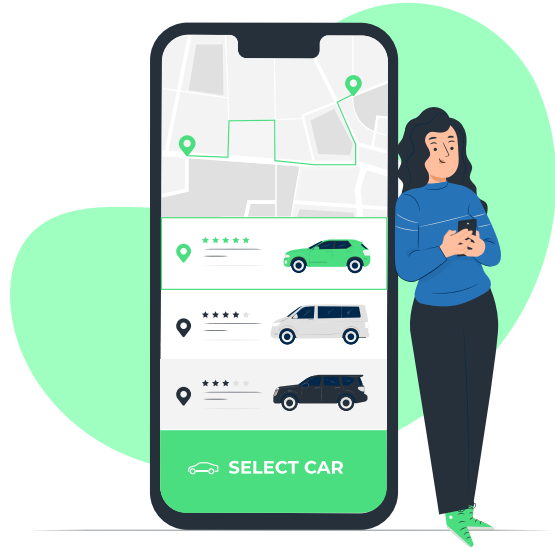
FLEETIX

Pitch Deck

"Where every route is the smart route!"



- DASS Team 28 -



Transform Your Employee Transportation with Intelligent Fleet Optimization

· 01 ·

**The Daily
Transportation
Challenge**

"Companies waste up to 30% of fleet resources through inefficient routing"

Are you facing :

- Unhappy employees due to long commutes?
- Vehicles running below capacity?
- Rising fuel costs from inefficient routes?
- Coordination nightmares with multiple vehicles?

Fleetix solves all these problems with one intelligent solution.

· 02 ·

The **FLEETIX** Advantage

Transforming Transportation



Optimised Routes

Reduce Travel
time



Cost Efficiency

Maximise Vehicle
Utilisation



Employee Satisfaction

Personalised
Pickup/Dropoff
Preferences



Sustainability

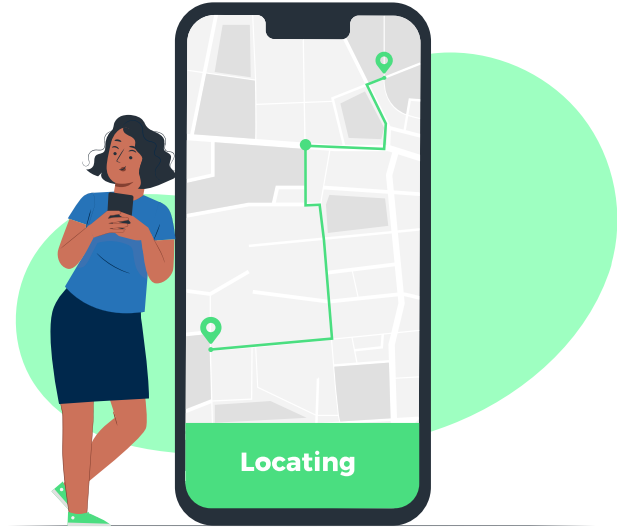
Lower carbon
footprint due to
efficient routing

- 03 -

Fleet Management **‘The Heart of Fleetix’**

Comprehensive Vehicle Management

- Manages multiple pickup and dropoff locations across your network
- Intelligently assigns appropriate vehicles to each route
- Balances fleet resources to maximize efficiency



Constraint Handling

C

Vehicle Capacity

Automatically balances passenger load and vehicle capacity across fleet

T

Shift Timings

Respects employee time preferences while optimizing fleet usage

D

Distance

Optimized for minimal total distance and time

M

Many-to-Many

Multiple vehicles servicing multiple employees across different pickups and dropoffs

- 04 -

**Experience the
Difference: User Roles**

Role-based Access

Admin	Employee	Driver
Fleet-wide visibility and control Fleet Resource allocation	Customizable preferences for pickup/dropoff	Turn-by-turn navigation with optimized stops
Performance analytics and report	Ride history and scheduling	Passenger details and special instructions

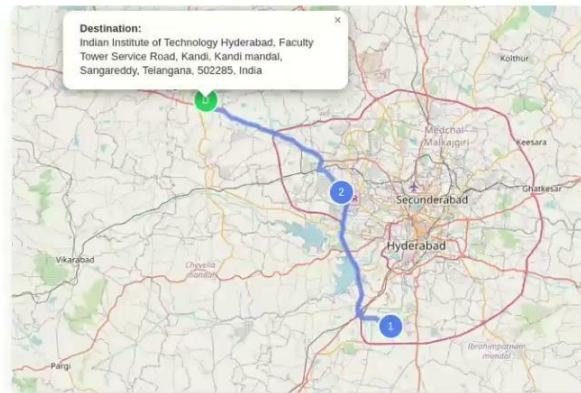
- 05 -

See Fleetix in Action:
Demo Video

Optimize your Employee Coordination

Efficient route optimization for multiple pickups and drop-offs using OpenStreetMap technology.

Say goodbye to manual route planning and hello to increased efficiency, reduced costs, and improved employee satisfaction.

[Learn More](#)[Request Demo](#)

- 06 -

R1 to R2: Our Evolution



R1

Foundation

- Geographic intelligence with **advanced geocoding**
- Core **routing** algorithms
- Intuitive **interface** design

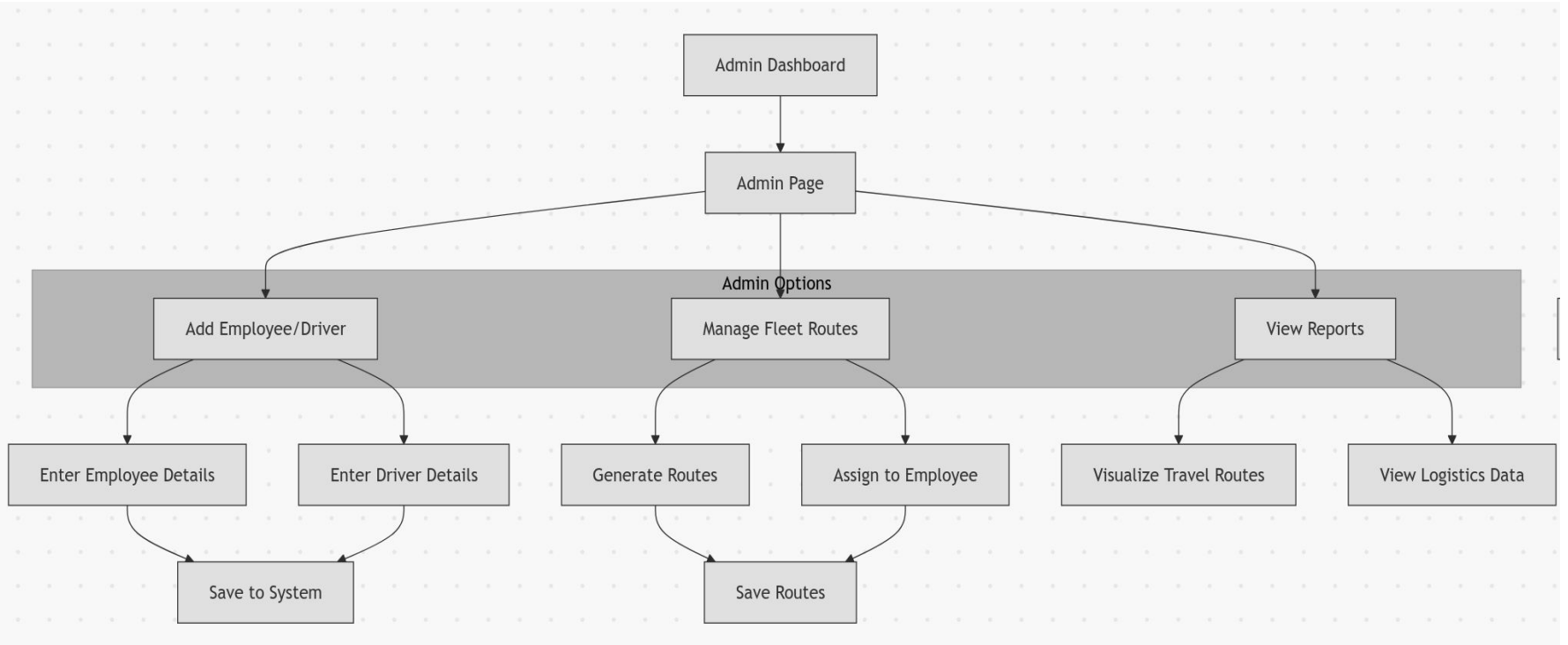


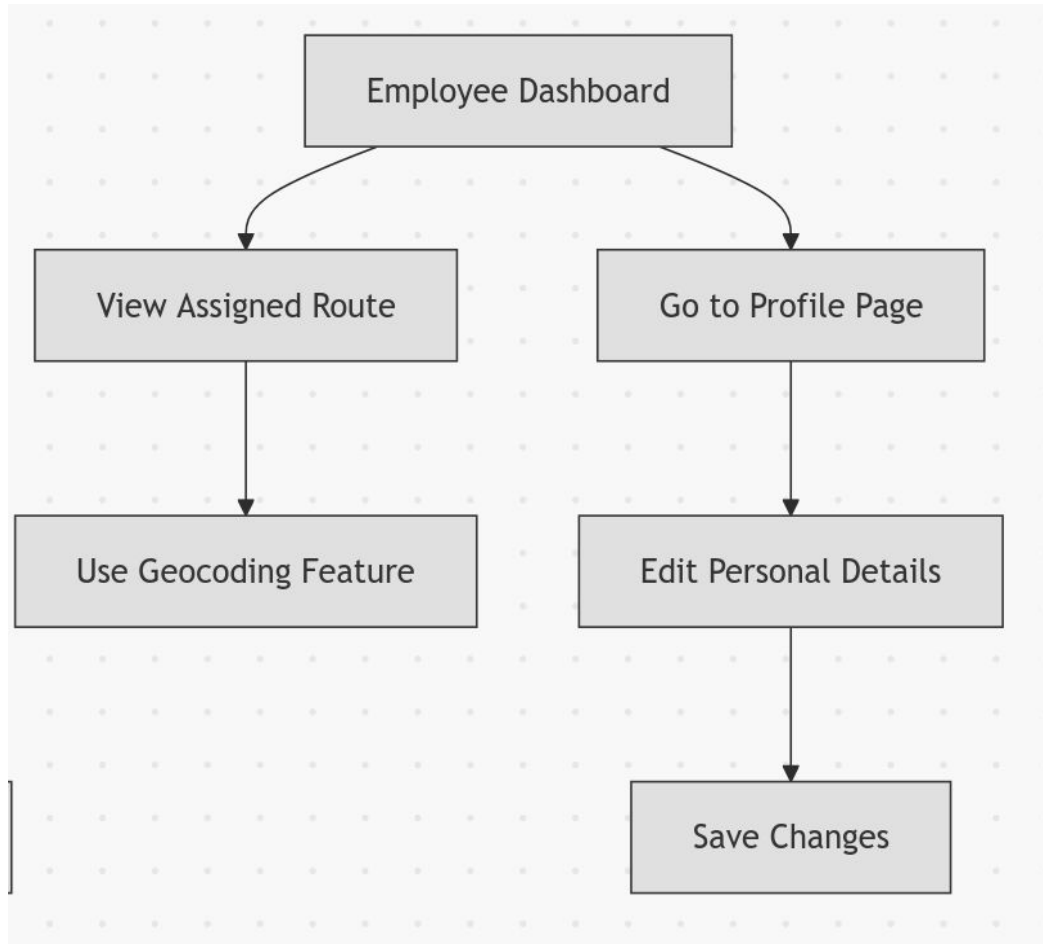
R2 Solution

- Full **fleet management** system
- **Constraint** -based routing optimization
- Role-based access with **personalized** dashboards

· 07 ·

System Overview





- 08 -

**Quality Attributes vs
Design Choices**



Scalability

Design Choice :

Separated fleet management
from routing logic

Impact:

System can scale to handle
thousands of vehicles without
query performance degradation



Usability

Design Choice:

Role-based UI architecture with
adaptive components

Impact:

Users only see relevant controls,
reducing cognitive load and
training time



Reliability

Design Choice :

Implemented driver assignment locks

Impact:

Prevents duplicate driver assignments and ensures each route has exactly one assigned driver



Security

Design Choice:

Role-based access control integrated with authentication

Impact:

Data segmentation ensures users only see their own transportation details

- 09 -

**Implementation
Journey : What We
Learned**

Technical Learnings



Constraint-based Optimisation

Implementing algorithms that balance multiple competing constraints in routing problems and fleet resource allocation



UI Design

Engineered responsive, multi-role dashboards featuring dynamic map rendering.



Database Design

Structuring data for efficient querying and updates

Non-technical Learnings



Project Management

Effective sprint planning and task prioritization



Agile Methodology

Adapting to changing requirements while maintaining progress



Team Collaboration

Coordinating interdependent components across team members



Documentation Practices

Creating living documents that evolve with the project

Future Works



Real-time updates

Incorporation of real-time traffic data and tracking of vehicles for better scheduling and time estimations



Driver Location

Considering the driver's current (or to be) locations for assigning their next trip



Driver Shifts

Implementing shift constraints for drivers to avoid overworking and uneven scheduling



Drop-off policies

Integrating safety policies and legal norms such as ensuring women aren't scheduled as the last drop off

Tech Stack

Full Stack

- **Next.js** framework for backend and frontend
- **TypeScript + Tailwind** CSS for frontend
- **PostgreSQL** for database

Routing Algos

- **OSM** (Open Street Maps) for map data
- **OSRM** (Open Source Routing Machine) for routing
- **K-Means Clustering** for can allocations

Hosted at : fleetix-team28.vercel.app

Work Distribution

Member	Main Role	Activities
Akshitha	Routing, Optimisation	Developed route optimization algorithms Handled route management backend integration
Aryanil	Research, Development Lead	Implemented admin and users UI Integrated backend for admin and users
Aviral	Coordinator, Frontend Lead	Implemented geocoding Developed complete user interface
Satvik	Backend, Database	Implemented database schema Designed admin analytics page
Sherley	Fleet allocation, Optimisation	Developed Fleet Allocation algorithms Implemented role-based route history

Acknowledgements

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- **Our Client, Akhil** , for his valuable feedback, and support in shaping the direction of our work.
- **Teaching Assistant, Ashna Dua** for her constant guidance and support throughout every phase of the project.

Thank You!

Thank You!

Any Questions, Feedbacks,
Suggestions would be appreciated!